EEE 3229 Communication Engineering-1 Part B Question Bank with Exam-Ready Answers

Lec 24–38 + Irfan sir's slides (Multiplexing, Multiple Access, Spread Spectrum)

What's inside: every Part-B-type question from the 2018-19, 2019-20, 2020-21 and 2021-22 finals, the old term tests, your own TT-3, and the slide/notes examples sir said to solve. They are arranged in lecture order so you can practise each one right after reading that lecture in your notes.
Marks key: ★★★ asked 3+ times · ★★ asked twice · ★ asked once or comes straight from sir's slides/notes. The [CO] tag shows which course outcome the question most likely tests.
Honest note: no guide can guarantee 100%, because the external examiner can add questions. This bank covers everything in the old Part-B papers plus the CO-mapped slide material. Learn the ★★★ and ★★ items word for word first.

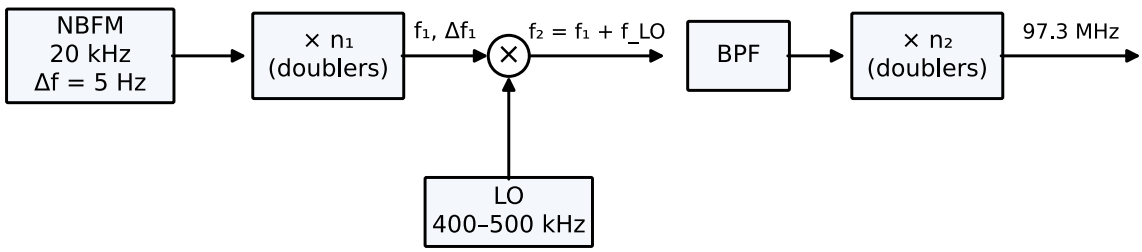
Notation: B or W = message bandwidth, fs = sampling rate, L = number of quantization levels, n = bits per sample, mp = peak message amplitude, σ = DM step size, Rb = bit rate.

Lec 24–27 · FM topics that have appeared in Part B

Old Part-B papers carried leftover FM questions (2018-19 Carson's rule, 2019-20 FM numerical + slope detection, 2020-21 Carson's rule), and your TT-3 was on Armstrong. Keep these in your Part B preparation too.

Q1. Design an Armstrong indirect FM modulator to generate an FM signal with carrier frequency 97.3 MHz and Δf = 10.24 kHz. An NBFM generator of fc1 = 20 kHz and Δf = 5 Hz is available. Only frequency doublers can be used as multipliers. A local oscillator (LO) adjustable between 400 and 500 kHz is available for frequency mixing.

★★★ 2019-20 Final, your TT-3 (20.48 kHz version) · [CO5] · your notes: "Armstrong must math in exam"



Total multiplication needed: n = Δf / Δf₁ = 10.24 kHz / 5 Hz = 2048 = 2^{11}. So we need 11 doublers in total: n_1 (before the mixer) × n_2 (after the mixer) = 2048.

The mixer changes only the carrier frequency, not Δf. After the first multiplier: f_1 = 20 kHz × n_1. The final carrier is 97.3 MHz, so the carrier just after the mixer must be

```
f₂ = 97.3 MHz / n₂ = 97.3 MHz × n₁ / 2048 = 47 509.77 × n₁  Hz
```

Using up-conversion ($f_2 = f_1 + f_{LO}$):

```
f_LO = f₂ − f₁ = (47 509.77 − 20 000) n₁ = 27 509.77 n₁
n₁ = 16  →  f_LO = 440.16 kHz   (lies within 400–500 kHz ✓)
```

($n_1 = 8$ gives 220 kHz and $n_1 = 32$ gives 880 kHz, both outside the LO range, so $n_1 = 16$ is the only valid choice.)

Stage	Carrier	Δf
NBFM output	20 kHz	5 Hz
× $n_1 = 16$ (4 doublers)	320 kHz	80 Hz
Mixer with f_LO = 440.156 kHz (sum taken by BPF)	760.156 kHz	80 Hz
× $n_2 = 128$ (7 doublers)	**97.3 MHz**	**10.24 kHz**

Answer: 4 doublers → mixer with LO = 440.16 kHz → 7 doublers.

TT-3 version (Δf = 20.48 kHz): $n = 20\,480/5 = 4096 = 2^{12}$. Then $f_2 = 97.3\text{ MHz} \times n_1/4096 = 23\,754.88\,n_1$, so $f_{LO} = f_2 - f_1 = 3\,754.88\,n_1$. $n_1 = 128$ gives $f_{LO} = 480.62$ kHz ✓. Chain: 20 kHz, 5 Hz → ×128 (7 doublers) → 2.56 MHz, 640 Hz → mix with 480.625 kHz → 3.040625 MHz, 640 Hz → ×32 (5 doublers) → **97.3 MHz, 20.48 kHz** ✓.

Q2. State Carson's rule and its limitations.

★★★ 2018-19 Final, 2020-21 Final, 2020-21 TT · [CO4]

Carson's rule: the bandwidth of an FM wave (tone or arbitrary message) is approximately

```
B_T ≈ 2(Δf + B) = 2B(β + 1)
β ≪ 1 (NBFM): B_T ≈ 2B          β ≫ 1 (WBFM): B_T ≈ 2Δf
```

Limitations:

1. It is an empirical (rule-of-thumb) formula with no exact proof.
2. It **underestimates** the bandwidth: it keeps only about 98% of the power. The 1% rule (count every sideband whose amplitude exceeds 1% of the unmodulated carrier) gives a bandwidth larger than $2(\Delta f + B)$, especially for $1 < \beta < 10$.
3. It is exact only for single-tone modulation. For a complex message it is only an estimate.
4. Theoretically FM has infinite sidebands ($J_n(\beta)$ is never exactly zero), so any finite bandwidth is an approximation.

Example (2020-21 TT): fm = 2000 Hz, Am = 5 V, k_f = 2 kHz/V → $\Delta f = 2 \times 5 = 10$ kHz, so $B_T = 2(10 + 2) =$ **24 kHz**.

Q3. In an FM system the frequency deviation is 5 kHz when the audio modulating frequency is 400 Hz and the modulating voltage is 2 V. Compute the modulation index. Also compute Δf and β if (i) the voltage is increased to 6 V at the same frequency, and (ii) the voltage is increased to 8 V while the frequency is reduced to 200 Hz.

★★ 2019-20 Final (Part B), 2020-21 Final (500 Hz / 2.5 V version) · [CO4]

$\Delta f \propto$ modulating voltage ($\Delta f = k_f \cdot Am$), so k_f = 5 kHz / 2 V = 2.5 kHz/V. Use $\beta = \Delta f / fm$.

```
Original: β = 5000/400 = 12.5
(i)  Am = 6 V: Δf = 2.5 × 6 = 15 kHz,  β = 15000/400 = 37.5
(ii) Am = 8 V, fm = 200 Hz: Δf = 2.5 × 8 = 20 kHz,  β = 20000/200 = 100
```

2020-21 version (500 Hz, 2.5 V → 5 kHz): k_f = 2 kHz/V. Original β = 10. At 7.5 V: Δf = 15 kHz, β = 30. At 10 V and 250 Hz: Δf = 20 kHz, β = 80.

Q4. Explain the principle of pre-emphasis and de-emphasis. Why are they necessary?

★★ 2018-19 Final, 2020-21 TT · Lec 27 · [CO3]

Why necessary: In FM, the noise power spectral density at the discriminator output rises as f^2 (a "triangular" noise spectrum), so noise is strongest at high audio frequencies. But in speech and music the high-frequency components (above about 2 kHz) have much smaller amplitude. So the high-frequency part of the message has a very poor SNR.

Pre-emphasis (at the transmitter, before modulation): a high-pass RC network (time constant 75 μs in broadcast FM, corner about 2.1 kHz) artificially boosts the high-frequency components of the message. They then modulate the carrier at a higher level, well above the noise.

De-emphasis (at the receiver, after demodulation): a low-pass RC network with the same time constant ($H_{de}(f) = 1/H_{pe}(f)$) attenuates the high frequencies by the same amount. This restores the original frequency balance of the message, and at the same time attenuates the high-frequency noise added in the channel.

Result: the message is undistorted ($H_{pe} \cdot H_{de} = 1$), while the output noise is greatly reduced. That gives an SNR improvement of about 10–13 dB in FM broadcasting at almost no cost.

Q5. Explain the slope detection method of FM demodulation.

★ 2019-20 Final (Part B) · Lec 25 · [CO3]

A slope detector converts FM into AM and then uses an envelope detector:

1. The FM signal is applied to an LC tuned circuit whose resonant frequency is slightly **off** the carrier frequency, so that fc lies on the linear (sloping) part of the resonance curve.
2. As the instantaneous frequency $f_i = f_c + k_f \cdot m(t)$ moves up and down the slope, the output amplitude of the tuned circuit changes in proportion to the frequency. FM is converted to an FM+AM signal whose envelope follows m(t). The tuned circuit acts as a differentiator: $d/dt[\cos(\omega_c t + \varphi(t))]$ gives an amplitude of $\omega_c + k_f \cdot m(t)$.
3. A diode envelope detector (diode + RC) then recovers m(t), and a DC block removes the constant term.

Drawbacks: the slope of a single tuned circuit is linear only over a small range, so there is distortion for large Δf. It also responds to amplitude noise, so a limiter is needed before it. The **balanced slope detector** (two tuned circuits, one above and one below fc, outputs subtracted) improves linearity.

Lec 27–28 · Sampling and pulse modulation

Q6. State and prove the Nyquist sampling theorem. Also find the interpolation formula for reconstructing an analog signal from its uniform samples.

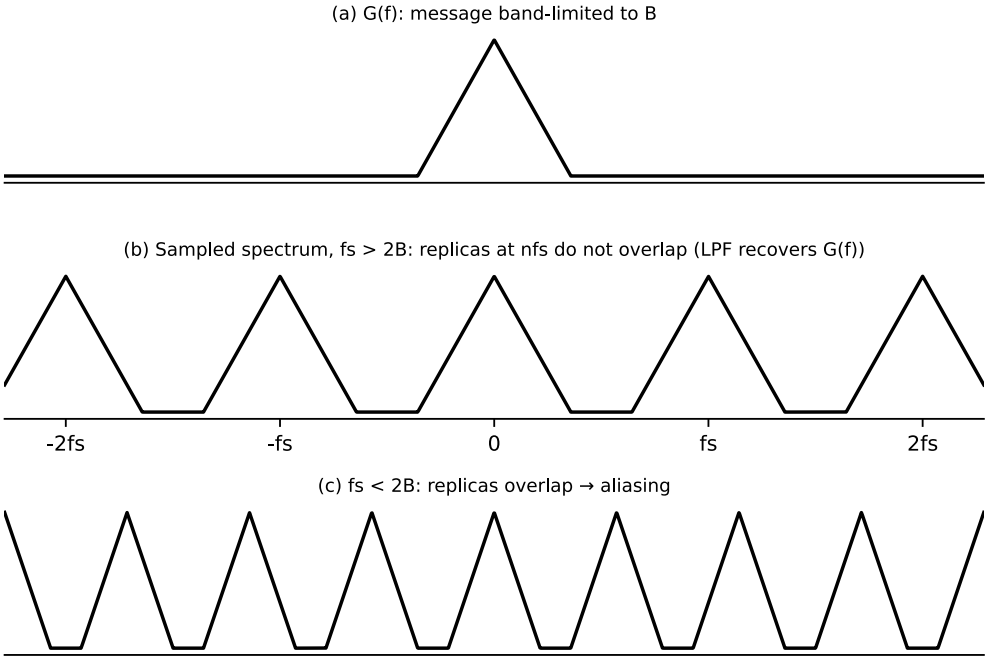
★★ 2020-21 Final, 2021-22 Final ("a proper LPF can reconstruct the sampled signal — justify") · [CO4]

Statement: A signal g(t) band-limited to B Hz ($G(f) = 0$ for $|f| > B$) can be reconstructed exactly from its samples taken uniformly at a rate $f_s \geq 2B$ samples/s (that is, $T_s \leq 1/2B$). The minimum rate $f_s = 2B$ is the Nyquist rate.

Proof: Ideal (instantaneous) sampling multiplies g(t) by an impulse train:

```
ḡ(t) = g(t)·δ_Ts(t) = Σₙ g(nTs) δ(t − nTs)
δ_Ts(t) is periodic, so as a Fourier series:  δ_Ts(t) = (1/Ts) Σₙ e^(j n ωs t),  ωs = 2π/Ts
⇒ ḡ(t) = (1/Ts) Σₙ g(t) e^(j n ωs t)
Using the frequency-shifting property  g(t)e^(j2πnfs t) ↔ G(f − nfs):
      Ḡ(f) = (1/Ts) Σₙ G(f − n fs)
```

So the spectrum of the sampled signal is G(f) repeated every fs Hz (scaled by 1/Ts). If fs ≥ 2B, the replicas do not overlap (figure b). An ideal LPF with cutoff B Hz and gain Ts then passes only the n = 0 term and gives back G(f) exactly, so g(t) is recovered. If fs < 2B, the replicas overlap (aliasing, figure c) and g(t) cannot be recovered. Hence fs ≥ 2B. ∎



Interpolation formula: take fs = 2B and the reconstruction filter H(f) = Ts·rect(f/2B):

```
h(t) = 2B·Ts·sinc(2πBt) = sinc(2πBt)        [sinc x = sin x / x]
g(t) = ḡ(t) * h(t) = Σₙ g(nTs) h(t − nTs)
     = Σₙ g(nTs) sinc[2πB(t − nTs)] = Σₙ g(nTs) sinc(2πBt − nπ)
```

Each sample is replaced by a sinc pulse centred at its sampling instant; the sum of these pulses is exactly g(t). This also justifies the statement "a proper LPF can reconstruct the sampled signal."

Q7. What is aliasing? How is it prevented? Name the types of sampling.

★ Lec 27–28 notes · [CO3]

Aliasing: when fs < 2B, or when the message has components above fs/2, the shifted spectra overlap. High-frequency components then appear ("fold back") as lower frequencies and distort the recovered signal.

Prevention: (i) pass the message through an **anti-aliasing LPF** with cutoff B before sampling, so nothing above B remains; (ii) sample slightly above the Nyquist rate (for example 20–25% above 2B) so that a practical, non-ideal reconstruction filter has a guard band.

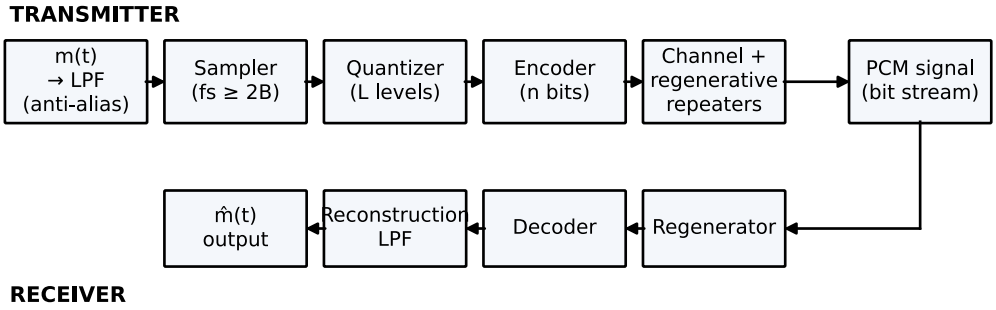
Types of sampling: (1) ideal/instantaneous sampling (impulse train); (2) natural sampling (gating: the pulse top follows g(t)); (3) flat-top sampling (sample-and-hold: constant pulse height = sample value). Flat-top samples give **PAM**. Other pulse modulations are PWM (width varies) and PPM (position varies).

Lec 29–32 · PCM, quantization, companding

Q8. What is PCM? Explain the PCM system with a block diagram. (also: Briefly explain the PCM steps.)

★★★ 2020-21 Final, 2021-22 Final, 2018-19 TT-2 · [CO3]

PCM (pulse code modulation) converts an analog message into a sequence of binary code words. The signal is sampled, each sample is quantized to one of L levels, and each level is coded with n = $\log_2 L$ bits.



1. **Anti-aliasing LPF:** band-limits m(t) to B Hz.
2. **Sampler:** takes samples at fs ≥ 2B (flat-top/PAM samples).
3. **Quantizer:** rounds each sample to the nearest of L levels (step Δv = 2mp/L). This introduces quantization error.
4. **Encoder:** represents each level by an n-bit binary code ($L = 2^n$), giving bit rate Rb = n·fs.
5. **Channel and regenerative repeaters:** along the line, repeaters detect and regenerate clean pulses, so noise does not accumulate.
6. **Receiver:** regenerator → decoder (bits back to quantized samples) → reconstruction LPF (cutoff B) gives $\hat{m}(t)$.

Advantages: high noise immunity, regeneration over long distances, easy TDM of many channels, encryption and error control are possible, and digital ICs are cheap.
Disadvantages: larger bandwidth (B_T = nB) and quantization noise.

Q9. Prove that for uniform quantization the signal-to-quantization-noise ratio is SQR = 3L²·P / mp². Hence show that (S/N)dB = 6.02n + 1.76 for a sinusoid.

★★ 2018-19 Final, Lec 30–31 · [CO2]

The message range (−mp, mp) is divided into L uniform steps:

$$\Delta v = \frac{2mp}{L}$$

The quantization error q = m − m̂ lies in (−Δv/2, Δv/2) and is equally likely anywhere in that range, so its pdf is p(q) = 1/Δv. The mean-square error (noise power) is

$$N_q = \bar{q}^2 = \int_{-\Delta v/2}^{\Delta v/2} q^2 \cdot \frac{1}{\Delta v}\, dq = \frac{\Delta v^2}{12} = \frac{(2mp/L)^2}{12} = \frac{mp^2}{3L^2}$$

With signal power $S_o = P = \bar{m}^2(t)$:

$$SQR = \frac{S_o}{N_q} = \frac{P}{mp^2/3L^2} = 3L^2 \cdot \frac{P}{mp^2} \qquad \blacksquare$$

In dB, for a sinusoid: $P = mp^2/2$, so SQR = $1.5L^2$. With $L = 2^n$:

$$(S/N)_{dB} = 10\log_{10}(1.5) + 10\log_{10}(2^{2n}) = 1.76 + 20n \cdot \log_{10}2 = 6.02n + 1.76 \text{ dB}$$

Conclusions: each extra bit adds about 6 dB of SQR. Since the PCM bandwidth B_T = nB, SQR grows exponentially with bandwidth (SNR ∝ 2^(2B_T/B)).

Q10. A sinusoidal wave is transmitted by PCM. If the SQR is required to be at least 47 dB, determine the minimum number of levels. Also find the SQR for this minimum value.

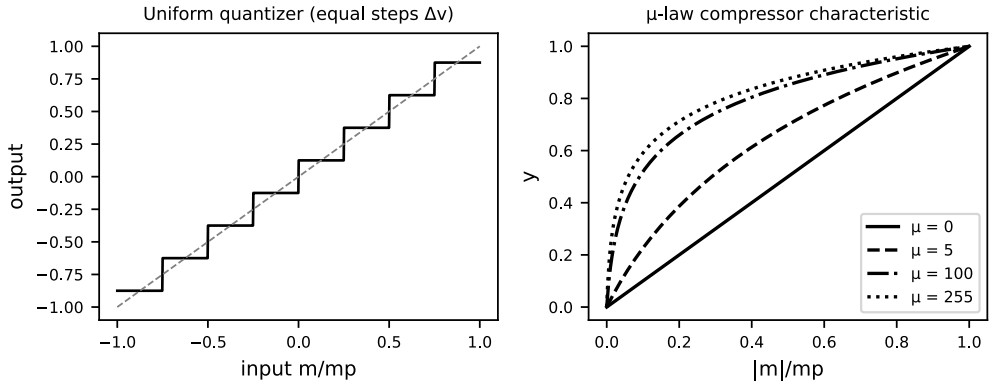
★ 2018-19 Final · [CO5]

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Sinusoid: SQR = 1.5 L²  ⇒  10log(1.5L²) ≥ 47
1.5L² ≥ 10^4.7 = 50 119  ⇒  L² ≥ 33 413  ⇒  L ≥ 182.8
Binary PCM needs L = 2ⁿ  ⇒  L = 256 (n = 8)
SQR = 1.76 + 6.02 × 8 = 49.92 dB  (≈ 50 dB)
```

(Using the formula directly: 6.02n + 1.76 ≥ 47 gives n ≥ 7.52, so n = 8 and L = 256. If the question does not require a power of 2, the minimum is L = 183.)

Q11. Define and explain uniform and non-uniform quantization. Write a short note on companding (μ-law and A-law).

★★★ 2018-19 Final, 2019-20 Final, 2021-22 Final (companding) · [CO3]



Uniform quantization: all steps are equal (Δv = 2mp/L). The noise power Δv²/12 is the same for weak and strong signals, so SQR = 3L²P/mp² is proportional to signal power. Weak signals (quiet talkers, the low-amplitude parts of speech, which are the most frequent) get a poor SQR. Since speech has a large ratio of peak to average power, uniform quantization wastes levels on rarely-used large amplitudes.

Non-uniform quantization: step sizes are small near zero and large at high amplitudes. Weak signals are quantized finely and strong signals coarsely, so the SQR stays nearly constant over a wide range of signal power (a large dynamic range) with the same number of bits.

Companding (compressing + expanding) is how non-uniform quantization is done in practice. A **compressor** (logarithmic amplifier) boosts small amplitudes before a **uniform** quantizer; at the receiver an **expander** with the inverse characteristic restores the original amplitudes. Compressor + uniform quantizer = non-uniform quantizer.

```
μ-law (N. America, Japan; μ = 255):
    y = sgn(m) · ln(1 + μ|m|/mp) / ln(1 + μ),                0 ≤ |m|/mp ≤ 1
A-law (Europe, Asia; A = 87.6):
    y = A(|m|/mp) / (1 + ln A),                               0 ≤ |m|/mp ≤ 1/A
    y = [1 + ln(A|m|/mp)] / (1 + ln A),                       1/A ≤ |m|/mp ≤ 1
With μ-law (μ ≫ 1):  SQR ≈ 3L² / [ln(1 + μ)]²  — nearly independent of signal power
```

Q12. A TV signal has a bandwidth of 4.5 MHz. It is sampled, quantized and binary coded to obtain a PCM signal. (i) Determine the sampling rate if the signal is sampled 20% above the Nyquist rate. (ii) If the samples are quantized into 1024

levels, find the number of bits per sample. (iii) Determine the binary pulse rate and the minimum bandwidth required.

★ 2018-19 Final · [CO5]

```
(i)   Nyquist rate = 2B = 9 MHz;  fs = 1.2 × 9 = 10.8 MHz (10.8 × 10⁶ samples/s)
(ii)  n = log₂1024 = 10 bits/sample
(iii) Rb = n·fs = 10 × 10.8 × 10⁶ = 108 Mbit/s
      Minimum bandwidth B_T = Rb/2 = 54 MHz
```

Q13. A television signal with a bandwidth of 4.2 MHz is transmitted using binary PCM with 512 quantization levels. Calculate (a) code word length, (b) transmission bandwidth, (c) final bit rate, (d) output SQNR.

★ old TT-2 · [CO5]

```
(a) n = log₂512 = 9 bits
(b) B_T = nW = 9 × 4.2 MHz = 37.8 MHz   (minimum, sampling at Nyquist rate)
(c) Rb = n·2W = 9 × 8.4 × 10⁶ = 75.6 Mbit/s
(d) SQNR = 1.76 + 6.02n = 1.76 + 54.18 = 55.94 dB  (sinusoidal signal assumed)
```

Q14. A central station is set up to monitor the ECGs of 10 hospital patients simultaneously. The data from the 10 patients are sampled, quantized, binary-coded and time-division multiplexed. The ECG bandwidth is 100 Hz. The maximum acceptable error in sample amplitude is 0.25% of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed.

★★★ 2020-21 Final, 2021-22 TT-3, 2021-22 Final · [CO5]

```
Nyquist rate = 2B = 200 Hz;  "twice the Nyquist rate" ⇒ fs = 2 × 200 = 400 samples/s
per patient
Maximum quantization error = Δv/2 = (2mp/L)/2 = mp/L
Requirement: mp/L ≤ 0.0025 mp  ⇒  L ≥ 1/0.0025 = 400
Binary: L = 2ⁿ ≥ 400  ⇒  L = 512,  n = 9 bits
Bit rate per patient = 400 × 9 = 3600 bit/s
Total (10 patients, TDM) = 10 × 3600 = 36 000 bit/s
Minimum cable bandwidth = Rb/2 = 36 000 / 2 = 18 kHz
```

If a version of the question says "sampled at the Nyquist rate" instead, use fs = 200 Hz, which halves the results to 18 kbit/s and 9 kHz.

Q15. Five telemetry signals, each of bandwidth 2400 Hz, are to be transmitted simultaneously by binary PCM. The signals must be sampled at least 20% above the Nyquist rate. Framing and synchronizing need an additional 0.5% extra bits. Determine the minimum possible data rate and the minimum bandwidth of the multiplexed signal.

★ 2019-20 Final · [CO5]

The paper did not state the accuracy. The original Lathi problem gives "maximum tolerable error in sample amplitudes = 0.2% of peak amplitude", so use that (if a different % or L is given, only n changes).

```
fs = 1.2 × 2B = 1.2 × 4800 = 5760 samples/s per signal
Error: mp/L ≤ 0.002 mp  ⇒  L ≥ 500  ⇒  L = 512, n = 9 bits
Bits/s per signal = 5760 × 9 = 51 840 bit/s
Five signals = 5 × 51 840 = 259 200 bit/s
+0.5% framing: Rb = 259 200 × 1.005 = 260 496 bit/s ≈ 260.5 kbit/s
Minimum bandwidth = Rb/2 ≈ 130.25 kHz
```

Q16. An analog signal of 4 kHz bandwidth is sampled at 1.25 times the Nyquist rate, and each sample is quantized into one of 256 equally likely levels. Assume the samples are statistically independent. (i) What is the information rate? (ii) Can the output be transmitted without error over an AWGN channel of bandwidth 10 kHz and S/N = 20 dB? (iii) Find the S/N required for error-free transmission in (ii). (iv) Find the bandwidth required for error-free transmission if S/N = 20 dB.

★ 2020-21 Final · [CO5]

```
fs = 1.25 × 2 × 4 kHz = 10 000 samples/s
Equally likely levels ⇒ H = log₂256 = 8 bits/sample
(i)   R = fs × H = 10 000 × 8 = 80 kbit/s
(ii)  S/N = 20 dB = 100.  C = B log₂(1 + S/N) = 10⁴ × log₂101 = 10⁴ × 6.658 = 66.58
kbit/s
      R = 80 kbit/s > C ⇒ NO, error-free transmission is not possible.
(iii) 80 × 10³ = 10⁴ log₂(1 + S/N) ⇒ 1 + S/N = 2⁸ ⇒ S/N = 255 (= 24.07 dB)
(iv)  B = R / log₂(1 + 100) = 80 000 / 6.658 = 12.02 kHz
```

Q17. An audio signal is sampled at 8 kHz using a 32-level quantizer. Find (i) the peak signal power to peak quantization noise power ratio, and (ii) the system bandwidth.

★ 2021-22 Final · [CO5]

```
L = 32 ⇒ n = 5 bits;   Δv = 2mp/L
Peak quantization error = Δv/2 = mp/L
(i)  Peak signal power / peak noise power = mp² / (mp/L)² = L² = 32² = 1024  (= 30.1
dB)
(ii) Rb = n·fs = 5 × 8000 = 40 kbit/s
     Minimum system bandwidth B_T = Rb/2 = 20 kHz
```

Q18. Write short notes on: Companding, Information rate, Entropy of a source.

★ 2021-22 Final (Part B) · [CO2]

Companding: see Q11. Compress the signal logarithmically before uniform quantization and expand it after decoding. This gives non-uniform quantization with a nearly constant SQR (μ-law μ = 255, A-law A = 87.6).

Entropy: the average information per message (symbol) of a source. For a memoryless source emitting $m_1 \ldots m_n$ with probabilities $P_1 \ldots P_n$:

```
H(m) = Σ Pᵢ log₂(1/Pᵢ) = −Σ Pᵢ log₂Pᵢ   bits/message
0 ≤ H ≤ log₂n; maximum when all messages are equally likely (maximum uncertainty)
```

Information rate: the average information produced per second. R = r·H bits/s, where r = number of messages per second. Example: probabilities 0.4, 0.3, 0.2, 0.1 → H = 1.846 bits; at $r = 10^6$ messages/s, R = 1.846 Mbit/s.

Q19. Explain differential PCM (DPCM). Why does it need fewer bits than PCM?

★ Lec 32–33 notes · [CO3]

Adjacent samples of a practical signal (speech, video) are highly correlated, so in PCM much of the information is redundant. In DPCM we transmit the **difference** between the current sample and its predicted value instead of the sample itself:

```
d[k] = m[k] − m̂[k]     (m̂[k] = predicted value from previous quantized samples)
Transmitter: predictor + quantizer in a feedback loop;  Receiver: m_q[k] = d_q[k] +
m̂[k]
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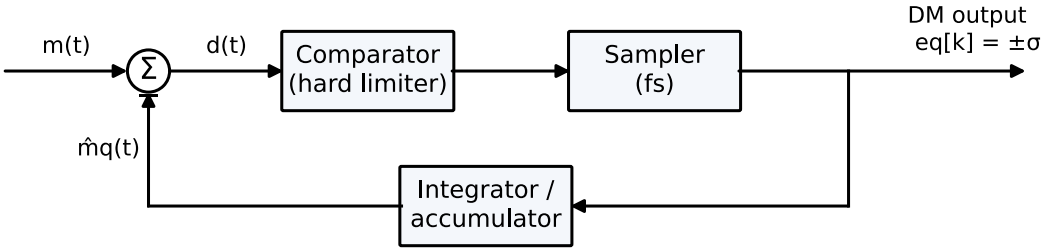
The difference d[k] has a much smaller peak value d_p than mp. For the same L, the step Δv = 2d_p/L is smaller, so the quantization noise (Δv²/12) is lower and the SNR improves. Equivalently, for the same SNR fewer bits per sample are needed, which reduces bit rate and bandwidth. (The simplest predictor m̂[k] = m_q[k−1] with a 1-bit quantizer gives delta modulation.)

Lec 33–35 · Delta modulation, ADM, Σ-Δ, line coding

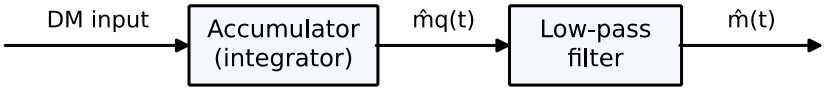
Q20. Draw the transmitter and receiver block diagrams of delta modulation. Show that the receiver of DM is essentially an accumulator.

★★ 2019-20 Final (accumulator), 2020-21 Final (block diagram) · [CO3]

(a) DM transmitter



(b) DM receiver



Principle: DM is one-bit DPCM. The message is compared with its staircase approximation m̂q(t). If m(t) > m̂q(t), a +σ pulse (bit 1) is sent; otherwise −σ (bit 0). The staircase moves one step σ up or down every Ts. It is sampled at many times the Nyquist rate so adjacent samples are close.

Receiver is an accumulator (proof): at the transmitter

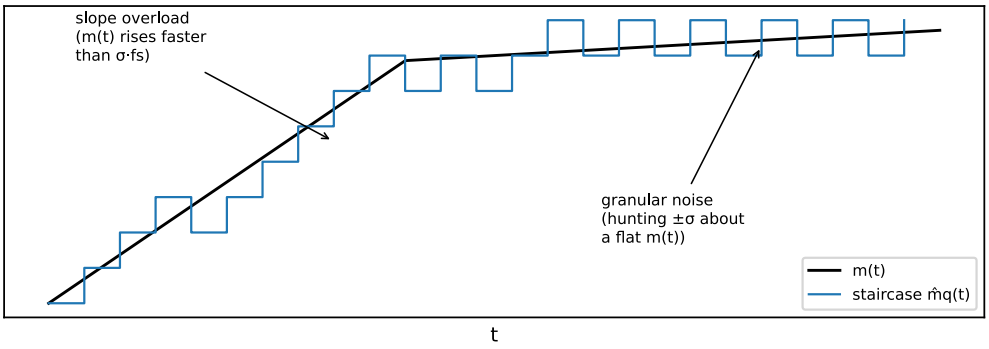
```
d[k]   = m[k] − m_q[k−1]
e_q[k] = σ·sgn(d[k]) = ±σ          (transmitted)
m_q[k] = m_q[k−1] + e_q[k]
Repeating the substitution:  m_q[k] = m_q[k−2] + e_q[k−1] + e_q[k] = … = Σ (i = 0 to k)
e_q[i]
```

So the estimate m_q[k] is just the running sum of all received pulses e_q[i]. The receiver only has to add up (accumulate/integrate) the incoming ±σ pulses, then smooth the staircase with an LPF. Hence the DM receiver is essentially an accumulator (integrator) followed by an LPF. ∎

Q21. Define slope overload distortion and granular noise in delta modulation. How

are they removed in ADM?

★★ 2020-21 Final, Lec 33–34 · [CO3]



Slope overload: the staircase can rise at most σ per Ts, that is with slope σ·fs. If the message changes faster than this, m̂q(t) cannot follow and falls behind, causing large distortion. To avoid it:

```
σ·fs ≥ |dm/dt|max ;   for m(t) = A cos ωm t:   A·ωm ≤ σ·fs
```

Granular noise: when m(t) is nearly constant, the staircase keeps hunting up and down by ±σ around it. This produces a noise similar to quantization noise, proportional to σ.

Conflict: a large σ reduces slope overload but increases granular noise; a small σ does the opposite.

Adaptive DM (ADM): the step size is varied according to the signal. When several successive bits are the same (1111 or 0000, a sign of slope overload) the step size is increased, for example multiplied by 1.5–2. When the bits alternate (1010, a sign of granular hunting) the step size is decreased. The receiver uses the same rule from the received bits, so no extra information is sent. ADM therefore reduces both slope overload and granular noise and gives a better SNR at the same bit rate.

Q22. A delta modulator is designed to operate at five times the Nyquist rate for a signal with 3 kHz bandwidth. The step size is 250 mV. Determine the maximum amplitude of a 2 kHz input sinusoid for which there is no slope overload.

★ 2018-19 Final · [CO5]

```
fs = 5 × (2 × 3 kHz) = 30 kHz;  σ = 0.25 V
No slope overload:  A·2πfm ≤ σ·fs
A_max = σ·fs / (2πfm) = (0.25 × 30 000) / (2π × 2000) = 7500 / 12 566 = 0.597 V
```

Q23. A DM system is tested with a 10 kHz sinusoid of 1 V peak-to-peak at the input. It is sampled at 10 times the Nyquist rate. Find the step size required to prevent slope overload.

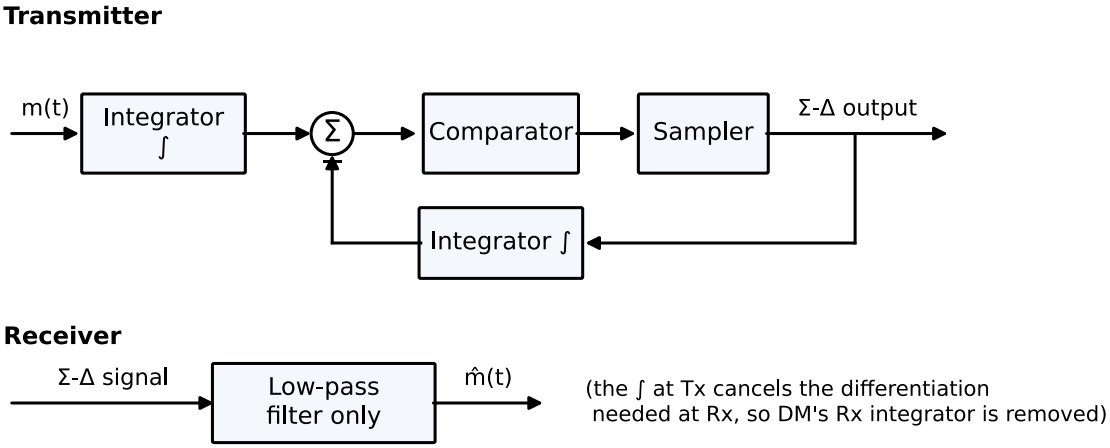
★ old TT-2 · [CO5]

```
A = 1/2 = 0.5 V;  fs = 10 × 2 × 10 kHz = 200 kHz
σ ≥ A·2πfm / fs = (0.5 × 2π × 10⁴) / (2 × 10⁵) = 31 416 / 200 000 = 0.157 V
Minimum step size ≈ 157 mV
```

Q24. Explain the principle of the sigma-delta modulator. Also explain how it is

advantageous over DM.

★★ 2019-20 Final, 2021-22 Final · [CO3]



Principle: in ordinary DM the receiver must integrate (accumulate) the pulses. In sigma-delta (Σ-Δ) modulation the message is **integrated first**, and the integrated signal ∫m(t)dt is delta-modulated. The two integrators (on the message path and on the feedback path) can be combined into a single integrator placed after the subtractor (comparing m(t) with the fed-back pulses).

At the receiver, DM would need an integrator to rebuild ∫m dt and then a differentiator to get m(t). The two cancel, so the **receiver is only a low-pass filter**.

Advantages over DM:

1. The receiver is simpler: just an LPF, with no accumulator.
2. Slope overload depends on the amplitude, not the slope, of m(t). Since ∫m dt changes slowly, high-frequency components are handled well, and DC (low-frequency) components can also be sent.
3. Pre-integration emphasizes the low-frequency content and increases the correlation between adjacent samples, which reduces the variance of the prediction error and improves SNR.
4. Channel noise is not accumulated at the receiver (no integrator there), so there is no build-up of noise.
5. With oversampling, quantization noise is pushed to high frequencies ("noise shaping") and removed by the LPF. This is why Σ-Δ is used in high-resolution ADCs and DACs.

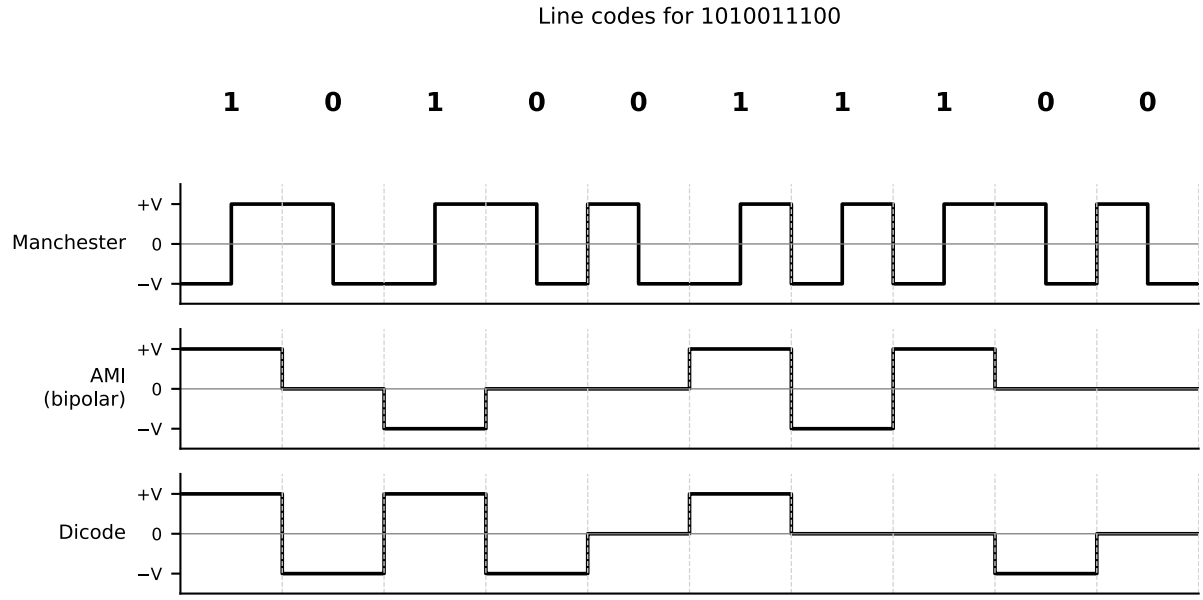
Q25. Briefly explain the desired characteristics of a line coding scheme. Draw the line-coded signal for 1010011100 for: (i) Manchester, (ii) AMI, (iii) Dicode.

★★ 2019-20 Final, 2018-19 Final (Q26) · Lec 35, Forouzan Ch 4 · [CO3/CO4]

Line coding converts a binary data sequence into a digital (pulse) signal for transmission. Desired properties:

1. **Small transmission bandwidth** (bandwidth efficiency).
2. **No DC component** and little low-frequency power, so that AC-coupled lines and transformers can be used (avoids baseline wandering).
3. **Self-synchronization / adequate timing content:** enough transitions for the receiver to recover the clock even for long runs of 0s or 1s.
4. **Power efficiency:** low transmitted power for a given error probability.
5. **Built-in error detection** (for example AMI: two successive 1s of the same polarity show an error).
6. **Transparency:** any data pattern is received correctly.

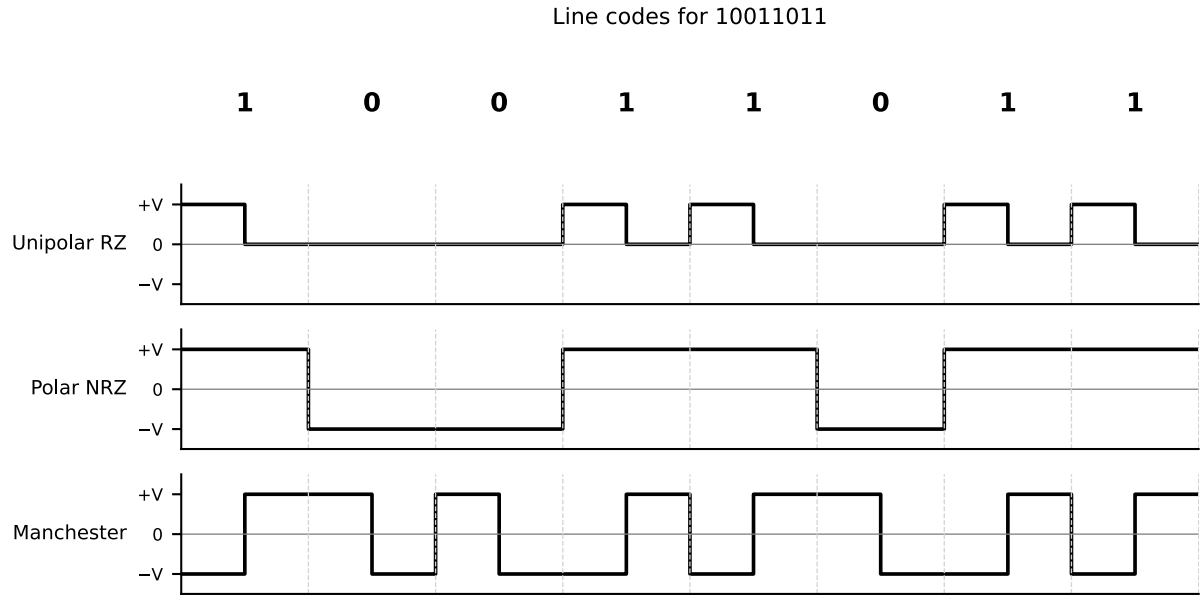
7. **Noise immunity** and **low complexity/cost**.



Conventions used (state them in the exam): **Manchester** (IEEE 802.3/Forouzan): 1 = low→high transition at mid-bit, 0 = high→low. **AMI (bipolar)**: 0 = zero level, 1 = alternately +V and −V. **Dicode**: output = present bit − previous bit (previous bit assumed 0): a 0→1 change gives +V, a 1→0 change gives −V, and no change gives 0. Levels for 1010011100: Dicode = +,−,+,−,0,+,0,0,−,0; AMI = +,0,−,0,0,+,−,+,0,0.

Q26. Show the graphical representation of the digital signal "10011011" represented as (i) unipolar RZ, (ii) polar NRZ, (iii) Manchester coding.

★ 2018-19 Final · [CO3]



Unipolar RZ: 1 = +V for the first half of the bit then 0; 0 = 0 for the whole bit. Polar NRZ: 1 = +V, 0 = −V for the full bit. Manchester: 1 = low→high, 0 = high→low.

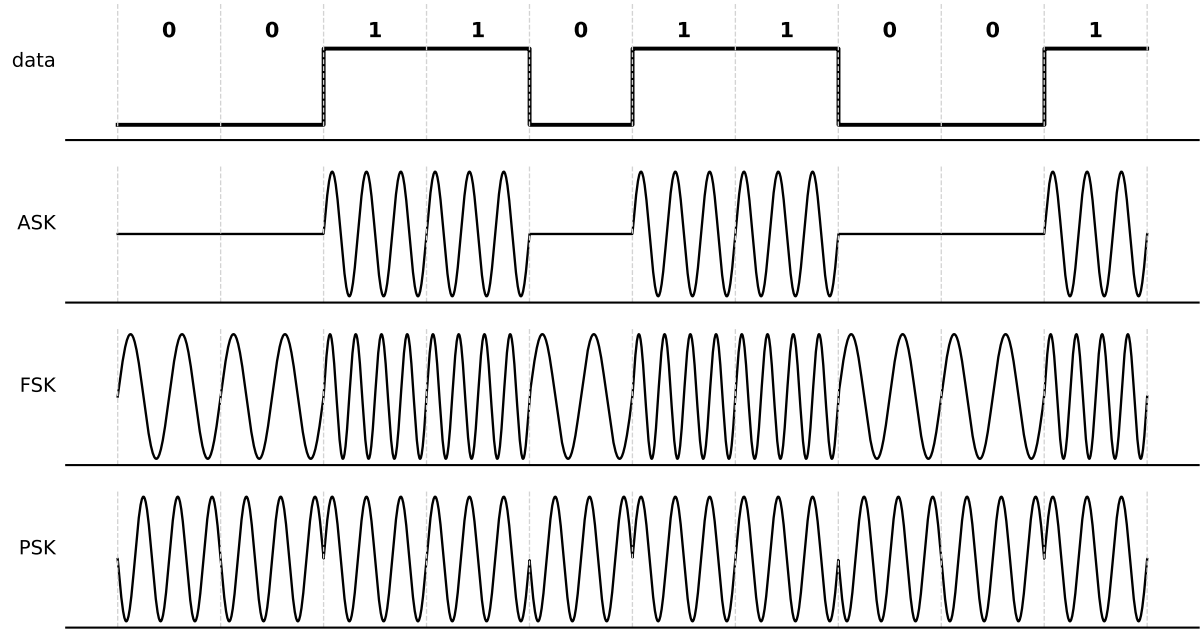
Lec 35–36 · Digital modulation: ASK, FSK, PSK, DPSK, QPSK, QAM

Q27. Name three mechanisms for modulating digital data into an analog signal.

Show the differences among them (draw for a bit sequence).

★★ 2021-22 Final, lab quiz (0011011001) · [CO3]

Digital-to-analog conversion changes one characteristic of a sinusoidal carrier according to the data: **ASK** (amplitude), **FSK** (frequency), **PSK** (phase).

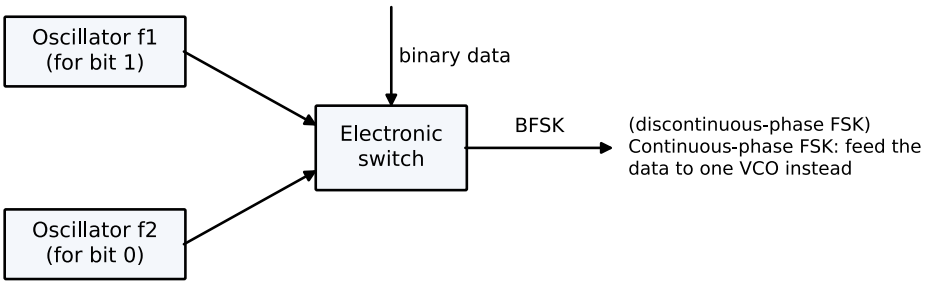


Feature	ASK (OOK)	FSK	PSK (BPSK)
What changes	Amplitude (1: A cos ωct, 0: 0)	Frequency (1: f_1, 0: f_2)	Phase (1: 0°, 0: 180°)
Bandwidth	(1 + d)·S ≈ 2Rb	(1 + d)·S + 2Δf (largest)	(1 + d)·S ≈ 2Rb
Noise immunity	Poorest (amplitude noise)	Better (constant envelope)	Best (lowest BER)
Detection	Coherent or envelope	Coherent or non-coherent	Coherent only (or DPSK)
Complexity	Simplest	Moderate	Most complex

S = signal (baud) rate, d = filtering factor (0 to 1).

Q28. Briefly describe FSK generation with the necessary block diagram. Include its spectrum and discuss the bandwidth requirements, along with the advantages and disadvantages of FSK.

★ 2020-21 Final · [CO3/CO4]



Generation: in binary FSK, bit 1 is sent as A cos $2\pi f_1 t$ and bit 0 as A cos $2\pi f_2 t$. (a) **Discontinuous-phase FSK:** two independent oscillators (f_1, f_2) are selected by an electronic

switch driven by the data; the phase may jump at bit boundaries. (b) **Continuous-phase FSK (CPFSK):** the polar NRZ data drives a single voltage-controlled oscillator (FM of the carrier by the data), so there is no phase jump. MSK is CPFSK with Δf = Rb/4, the minimum spacing for orthogonality.

Spectrum: BFSK can be viewed as two interleaved OOK (ASK) signals, one at carrier f_1 and one at f_2. Its spectrum is therefore two ASK spectra (sinc-shaped main lobes) centred at f_1 and f_2.

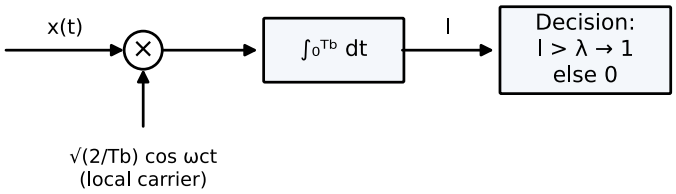
$$B_{FSK} \approx (1 + d)\cdot S + 2\Delta f \approx |f_1 - f_2| + 2Rb \qquad (\Delta f = |f_1 - f_2|/2)$$

Advantages: constant envelope, so it is less affected by amplitude noise and nonlinear amplifiers; simple non-coherent detection is possible; better noise immunity than ASK.
Disadvantages: requires more bandwidth than ASK/PSK (poor bandwidth efficiency); its BER is higher than PSK for the same Eb/N0; it is not suitable for high-speed links.

Q29. Briefly describe coherent and non-coherent detection of BASK and BFSK.

★★ 2019-20 Final (coherent BFSK), 2021-22 Final · [CO3]

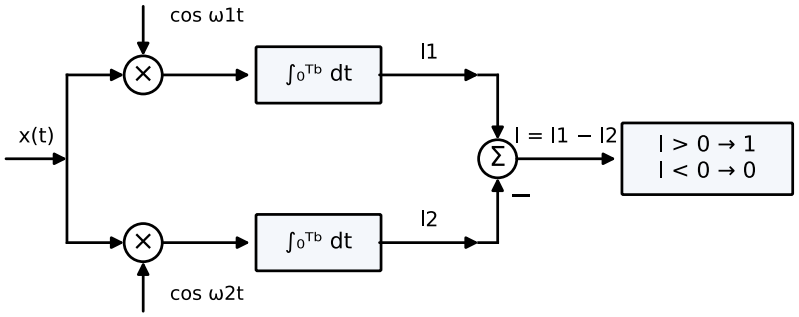
(a) Coherent BASK



(b) Non-coherent BASK



(c) Coherent BFSK



Non-coherent BFSK: replace each multiplier+integrator by a BPF (f1 / f2) followed by an envelope detector, then compare the two envelopes.

Coherent BASK: the received x(t) is multiplied by a locally generated carrier that is exactly synchronized in frequency and phase. The product is integrated over one bit (correlator), and the output l is compared with a threshold λ (half the expected "1" level): $l > \lambda \rightarrow 1$, otherwise 0.

Non-coherent BASK: no carrier phase is needed. A BPF centred at fc removes out-of-band noise, an envelope detector extracts the amplitude, and a decision device compares the envelope with a threshold.

Coherent BFSK: two correlators, with local carriers at f_1 and f_2. If 1 (f_1) was sent, l_1 is large and $l_2 \approx 0$ (orthogonal frequencies), and vice versa. The difference $l = l_1 - l_2$ is compared with zero: $l > 0 \rightarrow 1$, $l < 0 \rightarrow 0$.

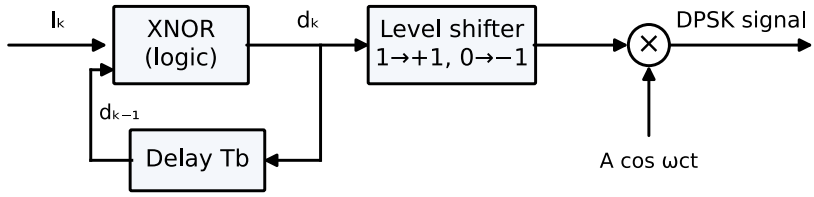
Non-coherent BFSK: two BPFs (at f_1 and f_2), each followed by an envelope detector. The decision chooses the branch with the larger envelope.

Coherent detection gives a lower BER but needs carrier recovery (complex); non-coherent detection is simpler but needs about 1–3 dB more signal power.

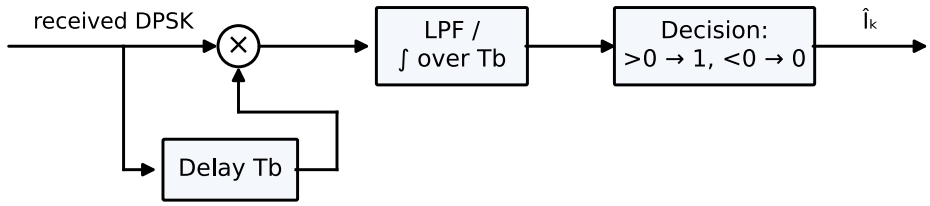
Q30. Explain the DPSK principle with a necessary block diagram.

★★ 2019-20 Final, your TT-3 (block diagram) · [CO3]

(a) DPSK transmitter



(b) DPSK receiver (no local carrier needed)



Principle: coherent PSK needs a local carrier with the exact phase, which is difficult to obtain. In DPSK the information is carried by the **phase change between successive bits**, not the absolute phase, so the receiver uses the previous received bit as its phase reference.

Transmitter: the data I_k is first differentially encoded:

```
dₖ = NOT(Iₖ ⊕ dₖ₋₁)     (XNOR; an extra reference bit d₀ = 1 starts the sequence)
Iₖ = 1 → dₖ = dₖ₋₁ (no phase change);   Iₖ = 0 → dₖ = complement of dₖ₋₁ (180° change)
```

d_k is converted to polar levels (1 → +1, 0 → −1) and BPSK-modulates the carrier: phase 0° for $d_k = 1$ and 180° for $d_k = 0$.

Receiver: the received signal is multiplied by itself delayed by one bit (Tb), then low-pass filtered (or integrated over Tb). If the two successive bits have the same phase the product is positive → 1; if their phases differ by 180° it is negative → 0. No local oscillator is needed.

Advantages: no carrier recovery circuit (simpler receiver). **Disadvantages:** about 1 dB poorer than coherent BPSK; an error in one bit tends to cause errors in two consecutive decisions.

Q31. Tabulate the values of the signal at each stage of the DPSK encoder and receiver for the message signal 1010011100.

★★★ 2018-19, 2020-21, 2021-22 (Q32), your TT-3 · [CO3] · Lathi Table

Rule: $d_k = \text{NOT}(I_k \oplus d_{k-1})$, reference bit $d_0 = 1$. Phase: $d_k = 1 \rightarrow 0$, $d_k = 0 \rightarrow \pi$. Receiver: same phase in two successive bits → 1; different phase → 0.

k	0	1	2	3	4	5	6	7	8	9	10
Input I_k	–	1	0	1	0	0	1	1	1	0	0
d_{k-1} (delayed)	–	1	1	0	0	1	0	0	0	0	1
Encoded d_k	1 (ref)	1	0	0	1	0	0	0	0	1	0
Polar level	+1	+1	−1	−1	+1	−1	−1	−1	−1	+1	−1
Transmitted phase	0	0	π	π	0	π	π	π	π	0	π
Rx: product of current × previous polarity	–	+	−	+	−	−	+	+	+	−	−
Detected output	–	1	0	1	0	0	1	1	1	0	0

The detected sequence 1010011100 equals the input ✓. (If your notes use a different reference bit or the XOR rule, the d_k row changes, but the method and the recovered output stay the same. State your rule first.)

Q32. Tabulate the values of the signal at each stage of the DPSK encoder and receiver for the message signal 1010010110.

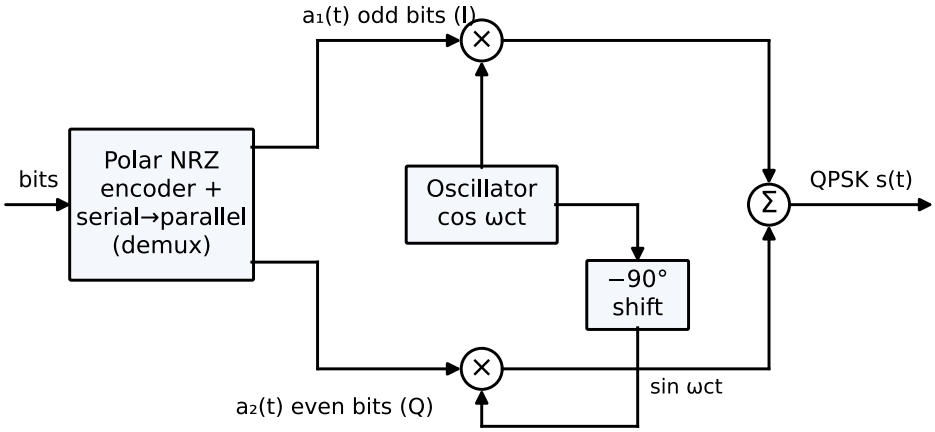
★ 2021-22 Final · [CO3]

k	0	1	2	3	4	5	6	7	8	9	10
Input I_k	–	1	0	1	0	0	1	0	1	1	0
d_{k-1}	–	1	1	0	0	1	0	0	1	1	1
Encoded d_k	1 (ref)	1	0	0	1	0	0	1	1	1	0
Transmitted phase	0	0	π	π	0	π	π	0	0	0	π
Rx product sign	–	+	−	+	−	−	+	−	+	+	−
Detected output	–	1	0	1	0	0	1	0	1	1	0

Output = 1010010110 ✓.

Q33. With a proper block diagram, describe the generation (modulation) of QPSK. Explain how it offers advantages over BPSK. Make an 8-PSK constellation diagram.

★★ 2018-19 Final, 2019-20 Final · Lec 36 · [CO3]



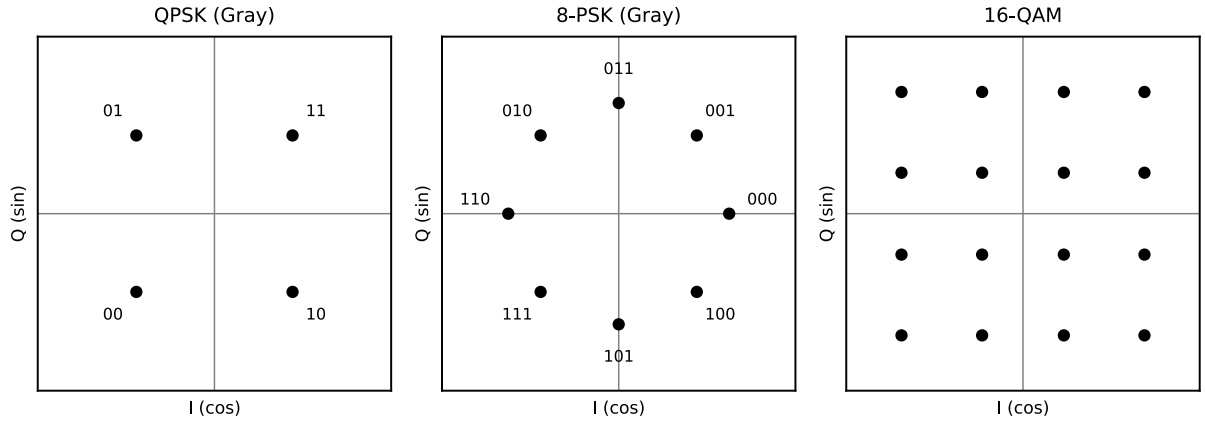
Generation:

1. The binary data is converted to polar NRZ (1 → +1, 0 → −1).

2. A serial-to-parallel converter (demultiplexer) splits it into two streams: odd-numbered bits $a_1(t)$ (in-phase, I) and even-numbered bits $a_2(t)$ (quadrature, Q). Each stream has half the original bit rate, so the symbol duration is T = 2Tb.
3. $a_1(t)$ multiplies cos ωct and $a_2(t)$ multiplies sin ωct (the carrier shifted by 90°). Each branch is a BPSK signal.
4. The two are added: $s(t) = a_1(t)\cos \omega ct + a_2(t)\sin \omega ct = \sqrt{2}\cdot A \cos(\omega ct - \theta)$. Each dibit gives one of four phases: 45°, 135°, 225°, 315°.

```
Gray-coded:  11 → 45°,  01 → 135°,  00 → 225°,  10 → 315°
```

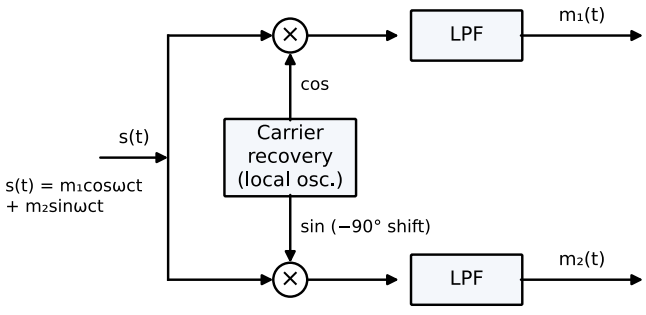
Advantages over BPSK: each symbol carries 2 bits (M = 4), so for the same bit rate QPSK needs **half the bandwidth** of BPSK; equivalently, it sends twice the data in the same bandwidth. Because the I and Q carriers are orthogonal, the bit error rate per bit (for the same Eb/N0) is the **same** as BPSK. So bandwidth efficiency doubles with no loss in error performance.



8-PSK: M = 8, 3 bits per symbol, 8 points equally spaced (45° apart) on a circle of radius A, Gray-coded so neighbours differ by one bit. 16-QAM: 4 bits per symbol; both amplitude and phase change.

Q34. With a proper block diagram, explain the principle of coherent detection of QAM. Mention its advantages and drawbacks.

★★ 2020-21 Final, 2021-22 Final · [CO3]



QAM signal: two independent messages modulate two carriers of the same frequency in quadrature: $s(t) = m_1(t)\cos \omega ct + m_2(t)\sin \omega ct$. Both occupy the same band.

Coherent detection: the receiver regenerates the carrier exactly (carrier recovery) and multiplies s(t) in two branches:

```
Upper: s(t)·2cos ωct = m₁(t) + m₁(t)cos 2ωct + m₂(t)sin 2ωct  →  LPF  →  m₁(t)
Lower: s(t)·2sin ωct = m₂(t) − m₂(t)cos 2ωct + m₁(t)sin 2ωct  →  LPF  →  m₂(t)
```

The LPFs remove the $2\omega c$ terms, and because cos and sin are orthogonal each branch recovers only its own message.

Advantages: two signals share one band, so bandwidth efficiency doubles (same bandwidth as one DSB signal). In digital form (M-QAM) many bits per symbol give very high data rates (used in modems, Wi-Fi, LTE, cable TV).

Drawbacks: requires exact carrier phase and frequency synchronization. With a phase error φ the outputs become $m_1\cos\varphi - m_2\sin\varphi$ (and similarly for the other branch), so each output has attenuation plus **co-channel interference (crosstalk)** from the other message; a frequency error makes this crosstalk vary with time. The receiver is complex, and dense M-QAM is sensitive to amplitude noise and needs linear amplifiers.

Lec 37 · Multiplexing: FDM, WDM, TDM (Forouzan Ch-6 slides)

Sir's note in Lec 37: "solve all the examples." All Ch-6 slide examples are included below.

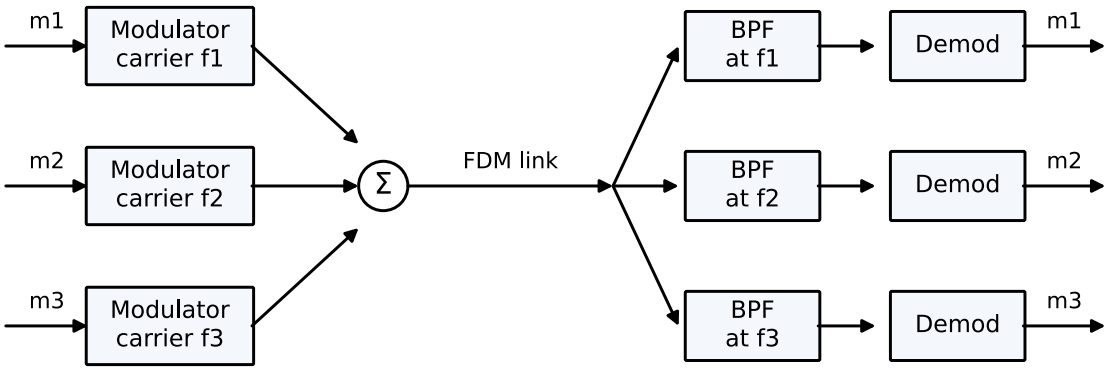
Q35. Differentiate between multiplexing and multiple access technologies.

★★ 2020-21 Final, 2021-22 Final · [CO3]

Multiplexing	Multiple access
Combines several signals at **one location** by a multiplexer (MUX) and sends them over one link; a DEMUX separates them at the other end.	Many **distributed users/stations** share a common channel (for example a radio band) directly, each accessing it on its own.
Resource sharing is controlled centrally by the MUX.	Sharing is controlled by an access protocol; each station uses its own band/slot/code.
Physical-layer technique, usually point-to-point links (trunks).	Data-link (MAC) layer channelization, point-to-multipoint (cellular, satellite).
Types: FDM, WDM, TDM.	Types: FDMA, TDMA, CDMA.
Example: telephone trunk carrying many calls; T-1 line.	Example: mobile phones in a cell sharing the base-station spectrum.

Q36. Briefly explain the frequency division multiplexing technique (with WDM).

★ 2019-20 Final · [CO3]



Ex 6.2: 5 × 100 kHz + 4 guard bands × 10 kHz = 540 kHz

FDM is an analog technique used when the link bandwidth is greater than the combined bandwidth of the signals. **Multiplexing:** each baseband signal modulates a different carrier frequency (f_1, f_2, f_3 …) so that the modulated signals occupy separate, non-overlapping bands. These are added (Σ) into one composite signal. **Guard bands** (unused strips) between channels prevent overlap/interference. **Demultiplexing:** a bank of band-pass filters

separates the channels, and each is demodulated to recover its message. Applications: AM/FM radio, TV broadcasting, the analog telephone hierarchy (group 12 → supergroup 60 → master group 600 → jumbo group 3600 voice channels), and first-generation cellular (AMPS).

WDM is the same idea for optical fibre: several light beams of different wavelengths are combined by a prism/diffraction grating into one fibre and separated again at the receiver. It is analog multiplexing of light signals.

Q37. FDM examples from the slides (Ex 6.1–6.4).

★★ Ex 6.2 asked in 2020-21 Final; sir: "solve all examples" · [CO5]

Ex 6.1: three 4 kHz voice channels into a 12 kHz link (20–32 kHz), no guard bands. Shift (modulate) channel 1 to 20–24 kHz, channel 2 to 24–28 kHz and channel 3 to 28–32 kHz, then combine. At the receiver, BPFs at 20–24, 24–28 and 28–32 kHz separate them, and each is demodulated back to 0–4 kHz.

Ex 6.2: five 100 kHz channels with 10 kHz guard bands. Five channels need at least four guard bands: B = 5 × 100 + 4 × 10 = **540 kHz**.

Ex 6.3: four 1 Mbps digital channels over a 1 MHz satellite channel using FDM. Divide into 4 channels of 1 MHz / 4 = 250 kHz each. Nyquist: $C = 2B \log_2 L$ → 1 Mbps = 2 × 250 k × $\log_2 L$ → $\log_2 L = 2$ → L = 4 (2 bits/signal element). So each 1 Mbps stream modulates a 250 kHz channel with **4-QAM**, and the four are frequency-shifted and combined (FDM).

Ex 6.4 (AMPS): each band 824–849 MHz and 869–894 MHz = 25 MHz. 25 MHz / 30 kHz = 833.33 → 832 channels in practice. Minus 42 control channels = **790 simultaneous users**.

Q38. Explain synchronous TDM, interleaving, synchronization (framing bits) and statistical TDM.

★ Lec 37–38, slides · [CO3]

TDM is a digital technique in which several low-rate channels share the time of one high-rate link. In **synchronous TDM** each input gets a fixed time slot in every frame, even if it has no data. If there are n inputs, a frame has n slots, the slot duration is 1/n of the input unit duration, the link rate is n × the input rate, and the **frame duration equals the input unit duration**.

Interleaving: the multiplexer, like a rotating switch, takes one unit (bit/byte/character) from each input in turn and places them in successive slots; the demultiplexer's switch rotates in synchronism to deliver each slot to the right output.

Synchronization: to know where each frame starts, one or more framing bits with a known pattern (for example alternating 1 0 1 0) are added at the beginning of each frame. The receiver locks onto this pattern and counts slots from it.

Inefficiency and statistical TDM: in synchronous TDM, slots of idle inputs go empty, wasting bandwidth. In **statistical TDM** slots are allocated dynamically only to inputs that have data, so the number of slots per frame is less than the number of inputs. Each slot must carry an **address** of its destination (overhead), and no framing synchronization bits are needed. It is more efficient but more complex.

Digital hierarchy (T-1): 24 voice channels × 8 bits + 1 framing bit = 193 bits/frame; 8000 frames/s → 193 × 8000 = **1.544 Mbps** (DS-1). DS-0 = 64 kbps, DS-2 = 6.312 Mbps (96 ch), DS-3 = 44.376 Mbps (672 ch), DS-4 = 274.176 Mbps (4032 ch). E-1 (Europe) = 2.048 Mbps (30 voice channels).

Q39. TDM examples 6.5–6.8 (quick answers).

★ slides; sir said solve all · [CO5]

Ex	Given	Answers
6.5	3 inputs × 1 kbps, unit = 1 bit	(a) input slot = 1 ms; (b) output slot = 1/3 ms; (c) frame = 3 × 1/3 = 1 ms
6.6	4 inputs × 1 Mbps, unit = 1 bit	(a) input bit = 1 µs; (b) output bit = 1/4 µs; (c) output rate = 4 Mbps; (d) frame rate = 1 000 000 frames/s
6.7	4 × 1 kbps, unit = 1 bit	(a) 1 ms; (b) link = 4 kbps; (c) slot = 250 µs; (d) frame = 1 ms
6.8	4 channels × 100 bytes/s, 1 byte per channel	frame = 4 bytes = 32 bits; frame rate = 100 frames/s; frame duration = 10 ms; link rate = 100 × 32 = 3200 bps

Q40. A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output for the inputs 110010, 001010, 101101, 000111. What is the frame rate? Frame duration? Bit rate? Bit duration?

★ 2019-20 Final (Ex 6.9) · [CO5]

Each frame takes 2 bits from each channel (channel 1 first; leftmost bit sent first):

Frame	Ch-1	Ch-2	Ch-3	Ch-4	Frame content
1	11	00	10	00	11 00 10 00
2	00	10	11	01	00 10 11 01
3	10	10	01	11	10 10 01 11

```
Output stream: 11001000 | 00101101 | 10100111
Bit rate of link = 4 × 100 kbps = 400 kbps
Frame size = 4 × 2 = 8 bits;  frame rate = 400 000 / 8 = 50 000 frames/s
Frame duration = 1/50 000 = 20 µs;   bit duration = 1/400 000 = 2.5 µs
```

Q41. We have four sources, each creating 250 characters (8 bits/character) per second. The interleaved unit is a character and 1 synchronizing bit is added to each frame. Find (a) the data rate of each source, (b) the duration of each character in each source, (c) the frame rate, (d) the duration of each frame, (e) the number of bits in each frame, and (f) the data rate of the link.

★★★ 2018-19 Final, 2021-22 TT-3 (Ex 6.10) · [CO5]

```
(a) 250 × 8 = 2000 bps = 2 kbps
(b) character duration = 1/250 s = 4 ms
(c) one character from each source per frame ⇒ frame rate = 250 frames/s
(d) frame duration = 1/250 = 4 ms  (same as one character duration)
(e) bits per frame = 4 × 8 + 1 = 33 bits
(f) link rate = 250 × 33 = 8250 bps
```

Q42. Two channels, one with a bit rate of 100 kbps and another with 200 kbps, are to be multiplexed. How can this be achieved? Frame rate? Frame duration? Bit rate of the link?

★ Ex 6.11 (multislot) · [CO5]

```
Multislot: give 1 slot to the 100 kbps channel and 2 slots to the 200 kbps channel.
Each frame = 3 bits;  frame rate = 100 000 frames/s  (1 bit from channel 1 per frame)
Frame duration = 1/100 000 = 10 µs;   link rate = 100 000 × 3 = 300 kbps
```

Q43. Two channels, one with a bit rate of 190 kbps and another with 180 kbps, are to be multiplexed using pulse-stuffing TDM with no synchronization bits. (a) Size of a frame in bits? (b) Frame rate? (c) Frame duration? (d) Data rate?

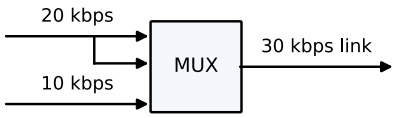
★ your TT-3 (likely to repeat) · [CO5]

```
Pulse stuffing: add 10 kbps of dummy bits to the 180 kbps channel ⇒ both at 190 kbps.
(a) 1 bit from each channel ⇒ frame = 2 bits
(b) frame rate = 190 000 frames/s
(c) frame duration = 1/190 000 = 5.26 µs
(d) data rate = 190 000 × 2 = 380 kbps
```

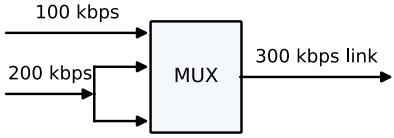
Q44. Write short notes on: (i) multilevel multiplexing, (ii) multiple-slot allocation, (iii) pulse stuffing.

★★★ 2019-20 Final, 2021-22 TT-3, 2021-22 Final · [CO3]

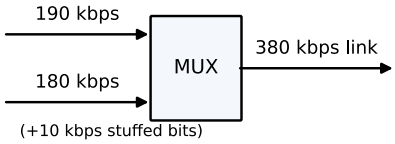
(a) Multilevel: 20 kbps input split into two 10 kbps lines → all inputs at 10 kbps



(b) Multislot: 100 kbps gets 1 slot, 200 kbps gets 2 slots per frame (Ex 6.11)



(c) Pulse stuffing: 180 kbps line raised to 190 kbps by inserting dummy bits



When the inputs of a TDM have different data rates, three strategies are used (data rate management):

(i) Multilevel multiplexing: used when the input rates are integer multiples of each other. A higher-rate input is split (demultiplexed) into several lines, each at the lowest common rate, and then all lines are multiplexed normally. Example: inputs 20, 10 and 10 kbps → the 20 kbps line becomes two 10 kbps lines, so four 10 kbps inputs give a 40 kbps link.

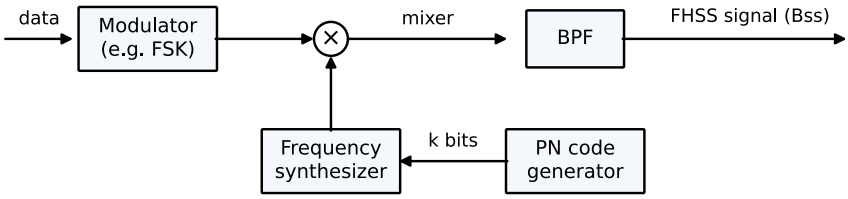
(ii) Multiple-slot allocation: used when the rates have a common factor (GCD). A higher-rate input is given more than one slot per frame, in proportion to its rate (the input is split by a serial-to-parallel converter into several slots). Example (Ex 6.11): 200 kbps gets 2 slots and 100 kbps gets 1 slot; frame = 3 slots, link = 300 kbps.

(iii) Pulse stuffing (bit padding / bit stuffing): used when the rates are not integer multiples. The slower input is raised to the rate of the faster input by inserting dummy bits, and the receiver removes them. Example: 180 kbps is stuffed to 190 kbps, then the two are multiplexed into a 380 kbps link.

Lec 38 + slides · Spread spectrum and multiple access (FDMA, TDMA, CDMA)

Q45. Explain the principle of the frequency hopping spread spectrum (FHSS) technique.

★★ 2018-19 Final, 2020-21 Final · [CO3]



Spread spectrum: a signal of bandwidth B is spread over a much larger bandwidth B_ss ≫ B by adding redundancy (a code), mainly to prevent **eavesdropping** and **jamming**. The spreading is independent of the data, and the receiver uses the same code to de-spread.

FHSS principle: the carrier frequency is changed ("hops") among M different frequencies according to a pseudo-random (PN) pattern. In each hopping period Th the signal occupies only one narrow band of width B, but over time it covers the whole B_ss.

1. A PN code generator produces a k-bit pattern every hopping period.
2. The frequency synthesizer converts each pattern into one of M = 2^k carrier frequencies (a frequency table).
3. The modulated data (for example FSK) is mixed with this frequency; a BPF keeps the wanted band.
4. The receiver uses an identical, synchronized PN generator and synthesizer to follow the hops and de-hop the signal back to its original band.

Example: k = 3 → 8 frequencies; the pattern sequence 101, 111, 001 … repeats after 8 hops (one cycle). **Bandwidth sharing:** if M frequencies are used, M users can share B_ss, each with a different hopping pattern (like FDM but with changing bands). **Advantages:** an eavesdropper without the code cannot follow the hops; a narrow-band jammer affects only a few hops; it also resists multipath and interference. Fast hopping = several hops per bit; slow hopping = several bits per hop.

Q46. Explain direct sequence spread spectrum (DSSS).

★ Spread spectrum slides, Lec 38 · [CO3]

In DSSS each data bit is replaced by n bits called **chips**, using a spreading code. The data is multiplied (XOR-ed) with the chip sequence, so the chip rate is n times the bit rate and the bandwidth is spread n times. Example: wireless LAN uses the 11-chip Barker sequence 10110111000; a data bit 1 is sent as the sequence and a 0 as its complement. The receiver multiplies by the same synchronized code (correlation) to recover the data; signals with other codes, noise and narrow-band jamming look like low-level noise after de-spreading. DSSS is the physical layer underlying **CDMA**.

Q47. Compare between (a) FDMA and CDMA, (b) TDMA and CDMA.

★★ 2021-22 Final (FDMA vs CDMA), 2020-21 Final (TDMA vs CDMA) · [CO3]

Feature	FDMA	TDMA	CDMA
Basic idea	Bandwidth divided into frequency bands; each station owns a band all the time	One channel time-shared; each station gets a time slot	Every station uses the whole bandwidth all the time, separated by unique orthogonal codes
Domain of separation	Frequency	Time	Code
Guard needed	Guard bands	Guard times + synchronization	None (orthogonal codes)
Transmission	Continuous	Burst (buffer needed)	Continuous, simultaneous
Capacity limit	Number of bands (hard)	Number of slots (hard)	Interference-limited (soft capacity)
Interference/security	Adjacent-channel interference; low security	Needs precise timing; moderate	Resistant to jamming, eavesdropping, multipath; high security
Handoff	Hard	Hard	Soft handoff possible
Complexity	Simple, analog	Moderate, digital	Complex (power control, code sync)
Example	1G AMPS	2G GSM, D-AMPS	IS-95, 3G (WCDMA)

Q48. Explain CDMA with chip sequences. Show how four stations share a channel and how a station recovers the data sent by another.

★ Multiple Access slides (Fig 12.24–12.28) · [CO3/CO5]

Rules of chip sequences (N = number of stations): each station is assigned a code of N chips. (1) Multiplying a sequence by a number multiplies every element. (2) The inner product of a sequence with itself = N. (3) The inner product of two different sequences = 0 (orthogonal). (4) Adding two sequences adds element by element.

Encoding: bit 0 → −1, bit 1 → +1, silent → 0. Each station multiplies its data by its code, and the channel carries the sum.

```
c₁ = [+1 +1 +1 +1]   c₂ = [+1 −1 +1 −1]   c₃ = [+1 +1 −1 −1]   c₄ = [+1 −1 −1 +1]
Station 1 sends 0 (d₁ = −1), station 2 sends 0 (d₂ = −1), station 3 silent (d₃ = 0),
station 4 sends 1 (d₄ = +1)
D = d₁c₁ + d₂c₂ + d₃c₃ + d₄c₄
  = [−1 −1 −1 −1] + [−1 +1 −1 +1] + [0 0 0 0] + [+1 −1 −1 +1] = [−1 −1 −3 +1]
```

Decoding (station 3 wants station 2's data): take the inner product of D with c_2 and divide by N:

```
D·c₂ = (−1)(+1) + (−1)(−1) + (−3)(+1) + (+1)(−1) = −1 + 1 − 3 − 1 = −4
d₂ = −4/4 = −1  ⇒  bit 0 ✓
Check station 4: D·c₄ = −1 + 1 + 3 + 1 = 4 ⇒ +1 ⇒ bit 1 ✓;  station 3: D·c₃ = 0 ⇒
silent ✓
```

Q49. Prove that a receiving station can get the data sent by a specific sender if it multiplies the entire data on the channel by the sender's chip code and then divides it by the number of stations.

★ Example 12.8 (slides) · [CO3]

For the four-station system, the data on the channel is $D = d_1c_1 + d_2c_2 + d_3c_3 + d_4c_4$. To get station 1's data, multiply by c_1:

```
D·c₁ = (d₁c₁ + d₂c₂ + d₃c₃ + d₄c₄)·c₁
     = d₁(c₁·c₁) + d₂(c₂·c₁) + d₃(c₃·c₁) + d₄(c₄·c₁)
     = d₁ × N + d₂ × 0 + d₃ × 0 + d₄ × 0          (c·c = N, different codes orthogonal
→ 0)
     = d₁ × N
⇒  (D·c₁) / N = d₁        ∎
```

Q50. How are chip sequences generated using Walsh tables? Find the chips for a network with (a) two stations, (b) four stations. What is the number of sequences if we have 90 stations?

★ Examples 12.6, 12.7 (slides) · [CO5]

```
Rules:  W₁ = [+1]          W₂ₙ = | Wₙ   Wₙ  |
                                 | Wₙ  −Wₙ  |   (−Wₙ = complement of Wₙ)
W₂ = | +1 +1 |        W₄ = | +1 +1 +1 +1 |
     | +1 −1 |              | +1 −1 +1 −1 |
                           | +1 +1 −1 −1 |
                           | +1 −1 −1 +1 |
```

Each row is the chip sequence of one station. (a) Two stations: [+1 +1] and [+1 −1]. (b) Four stations: [+1 +1 +1 +1], [+1 −1 +1 −1], [+1 +1 −1 −1], [+1 −1 −1 +1].

Ex 12.7: the number of sequences in a Walsh table must be N = 2^m. For 90 stations: $2^6 = 64 < 90 \le 2^7 = 128 \rightarrow m = 7$, **N = 128**. We use 90 of the 128 sequences as chips.

Communication system design (CO5)

Q51. Discuss the design procedure of a communication system.

★ 2021-22 Final · syllabus: design parameters, channel selection criteria, performance simulation · [CO5]

1. **Define requirements (design parameters):** type of message (voice, video, data), message bandwidth or data rate, required quality (output SNR or BER), range/distance, number of users, mobility, power and cost limits.
2. **Select the channel (channel selection criteria):** available bandwidth, attenuation per km, noise and interference, distance, data rate, security, mobility and cost. Example: twisted pair for short low-rate links, coaxial cable for medium bandwidth, optical fibre for very high rate and long distance (low loss, EMI immunity), and radio (with the right band and propagation mode) for wireless/mobile links.
3. **Check feasibility with channel capacity:** $C = B \log_2(1 + S/N)$ must exceed the required information rate; find the minimum SNR or bandwidth.
4. **Choose source coding and modulation:** analog (AM/DSB/SSB/VSB/FM) or digital (PCM/DM + ASK/FSK/PSK/QPSK/QAM), trading bandwidth against power/SNR. Choose multiplexing/multiple access (FDM/TDM, FDMA/TDMA/CDMA) for many users.
5. **Link budget:** transmitter power, antenna gains, path loss, noise figure/noise temperature of the receiver, required received SNR, fade margin.
6. **Design the transmitter and receiver blocks** (filters, modulators, detectors, synchronization, error control).

7. **Performance simulation:** simulate the complete system (for example in MATLAB/Simulink) to verify spectrum, SNR, BER and distortion under realistic noise and channel conditions; optimize the parameters.
8. **Prototype, test and iterate** until the specifications are met at minimum cost.

Last-night formula sheet (Part B)

```
Sampling:      fs ≥ 2B;  G̃(f) = (1/Ts) Σ G(f − nfs);  g(t) = Σ g(nTs) sinc(2πBt − nπ)
PCM:           L = 2ⁿ;  Δv = 2mp/L;  Nq = Δv²/12 = mp²/(3L²);  Rb = n·fs;  B_T(min) =
Rb/2 = nB
SQNR:          3L²·m̄²/mp²;  sinusoid → 1.5L² → 6.02n + 1.76 dB  (+6 dB per bit)
Max error:     Δv/2 = mp/L  (use for "error ≤ x% of peak" problems; ECG → L = 512, 36
kbps, 18 kHz)
μ-law:         y = ln(1 + μ|m|/mp)/ln(1 + μ),  μ = 255;  A-law A = 87.6
DM:            no slope overload: σ·fs ≥ |dm/dt|max;  sinusoid: A·2πfm ≤ σ·fs
Shannon:       C = B log₂(1 + S/N);  Nyquist (noiseless): C = 2B log₂L
FM:            β = Δf/fm;  Carson B_T = 2(Δf + B);  Armstrong: n₁·n₂ = Δf_final/Δf_NBFM
TDM:           frame duration = input unit duration;  link rate = n × input rate (+ sync
bits)
T-1:           (24 × 8 + 1) × 8000 = 1.544 Mbps
FDM:           B = N·B_ch + (N − 1)·guard
CDMA:          N = 2^m Walsh codes;  d_i = (D·c_i)/N
DPSK:          dₖ = NOT(Iₖ ⊕ dₖ₋₁), ref d₀ = 1;  Rx: same phase → 1, change → 0
QPSK:          2 bits/symbol, half the BPSK bandwidth, same BER
```